sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in **ggplot2** plots, **gt** and **reactable** tables, and **plotly**, **ggiraph** and **leaflet**, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ **COOKBOOKS ↗**

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: Getting started ↗

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: Logos, wordmarks, headshots ↗

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE` `HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL` `SD` `OAK`; `SA` `TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: Team colours and themes ↗

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: Logo axes and titles ↗

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

sdvplotR v0.3.0

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ **COOKBOOKS ↗**

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league. **Football** `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for **gt**, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0
gt ≥ 0.10 · paletteer optional
GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()` : page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)` : `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

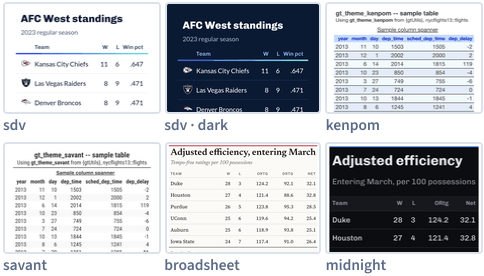
EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```



`gt_theme_sdv()` : Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"` : on the SDV navy.

`gt_theme_sdv_team()` : the sdv table in one team's colours (title primary, line secondary); `team = NULL` : SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)` : `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight` : a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paleteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info (label = colour)`, `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition (~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for **gt**: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the **reactable** helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column `to − from`: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "–"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: Rolling-window wins ↗

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars_top()` / `_bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: Border bars ↗

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column_subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: Headers, legends, captions ↗

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: Faceted tables ↗ · Schedule matrix ↗

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir, match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: Saving and posting ↗ · Social posting ↗

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv_logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv_wordmarks()`	the same
`reactable_sdv_headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv_team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv_team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv_cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv_theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: reactable integration ↗

SIBLINGS

Python **sdvplot** (PyPI) and JavaScript **@sportsdataverse/sdvplot** + **sdvtables** (npm) share sdvplotR's team index, logo archive and argument names; images render through **ggpath**.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in **ggplot2** plots, **gt** and **reactable** tables, and **plotly**, **ggiraph** and **leaflet**, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: Getting started ↗

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: Logos, wordmarks, headshots ↗

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE` `HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL` `SD` `OAK`; `SA` `TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: Team colours and themes ↗

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: Logo axes and titles ↗

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

sdvplotR v0.3.0

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league. **Football** `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x` `court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x` `pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for **gt**, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0
gt ≥ 0.10 · paletteer optional
GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()`: page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)`: `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

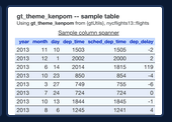
```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```

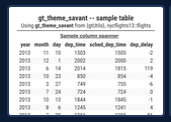


sdv



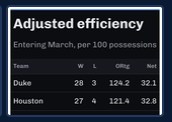
sdv · dark


kenpom


savant



broadsheet



midnight

`gt_theme_sdv()`: Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"`: on the SDV navy.

`gt_theme_sdv_team()`: the sdv table in one team's colours (title primary, line secondary); `team = NULL`: SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)`: `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight`: a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paletteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info` (`label = colour`), `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for **gt**: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the **reactable** helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column to `- from`: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "-"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: **Rolling-window wins ↗**

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars_top()` / `_bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: **Border bars ↗**

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column_subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: **Headers, legends, captions ↗**

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: **Faceted tables ↗** · **Schedule matrix ↗**

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data`, `group`, `fn`, `file`, `dir`, `match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: **Saving and posting ↗** · **Social posting ↗**

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv_logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv_wordmarks()`	the same
`reactable_sdv_headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv_team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv_team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv_cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv_theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: **reactable integration ↗**

SIBLINGS

Python **sdvplot** (PyPI) and JavaScript **@sportsdataverse/sdvplot** + **sdvtables** (npm) share sdvplotR's team index, logo archive and argument names; images render through **ggpath**.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in ggplot2 plots, gt and reactable tables, and plotly, ggiraph and leaflet, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: Getting started ↗

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: Logos, wordmarks, headshots ↗

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE` `HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL` `SD` `OAK`; `SA` `TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: Team colours and themes ↗

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: Logo axes and titles ↗

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

`sdvplotR v0.3.0`

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league. **Football** `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada. whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for gt, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0

gt ≥ 0.10 · paletteer optional

GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()` : page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)` : `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

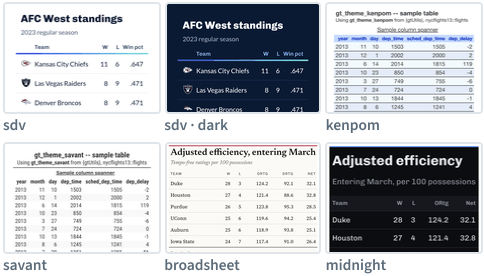
EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```



sdv · sdv · dark · kenpom · savant · broadsheet · midnight

`gt_theme_sdv()` : Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"` : on the SDV navy.

`gt_theme_sdv_team()` : the sdv table in one team's colours (title primary, line secondary); `team = NULL` : SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)` : `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight` : a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paleteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info (label = colour)`, `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for **gt**: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the **reactable** helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column `to - from`: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "-"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: Rolling-window wins ↗

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars _top() / _bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: Border bars ↗

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column _subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: Headers, legends, captions ↗

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: Faceted tables ↗ · Schedule matrix ↗

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir, match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: Saving and posting ↗ · Social posting ↗

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv _logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv _wordmarks()`	the same
`reactable_sdv _headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv _team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv _team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv _cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv _theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: reactable integration ↗

SIBLINGS

Python **sdvplot** (PyPI) and JavaScript **@sportsdataverse/sdvplot** + **sdvtables** (npm) share sdvplotR's team index, logo archive and argument names; images render through **ggpath**.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in ggplot2 plots, gt and reactable tables, and plotly, ggiraph and leaflet, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: Getting started ↗

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: Logos, wordmarks, headshots ↗

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE` `HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL` `SD` `OAK`; `SA` `TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: Team colours and themes ↗

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: Logo axes and titles ↗

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on plotly, ggiraph and leaflet, premade tier plots, 59 playing surfaces drawn with sportyR (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

sdvplotR v0.3.0

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`.
`sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded.
`sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league.
Football `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for gt, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0
gt ≥ 0.10 · paletteer optional
GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()`: page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)`: `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

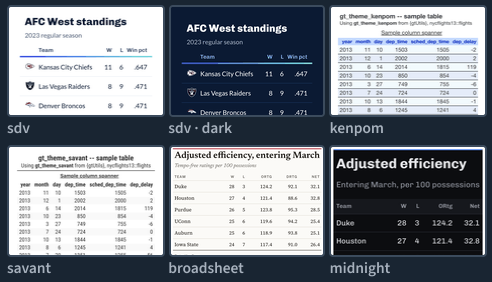
EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```



sdv · sdv · dark · kenpom · savant · broadsheet · midnight

`gt_theme_sdv()`: Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"`: on the SDV navy.

`gt_theme_sdv_team()`: the sdv table in one team's colours (title primary, line secondary); `team = NULL`: SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet paper = `"white"` or `"salmon"` · almanac stripe (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)`: `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight`: a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paletteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info` (`label = colour`), `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for gt: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the reactable helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column to − from: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "–"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: Rolling-window wins ↗

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars_top()` / `_bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: Border bars ↗

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column_subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: Headers, legends, captions ↗

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: Faceted tables ↗ · Schedule matrix ↗

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir, match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: Saving and posting ↗ · Social posting ↗

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv_logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv_wordmarks()`	the same
`reactable_sdv_headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv_team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv_team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv_cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv_theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: reactable integration ↗

SIBLINGS

Python sdvplot (PyPI) and JavaScript @sportsdataverse/sdvplot + sdvtables (npm) share sdvplotR's team index, logo archive and argument names; images render through ggpath.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in **ggplot2** plots, **gt** and **reactable** tables, and **plotly**, **ggiraph** and **leaflet**, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: **Getting started ↗**

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: **Logos, wordmarks, headshots ↗**

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL SD OAK`; `SA TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: **Team colours and themes ↗**

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: **Logo axes and titles ↗**

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

`sdvplotR v0.3.0`

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league. **Football** `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for **gt**, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0

gt ≥ 0.10 · paletteer optional

GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()` : page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)` : `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

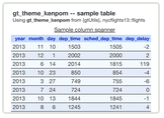
```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```

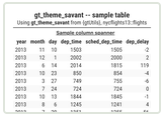


sdv



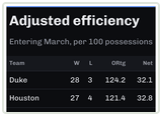
sdv · dark


kenpom


savant



broadsheet



midnight

`gt_theme_sdv()` : Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"` : on the SDV navy.

`gt_theme_sdv_team()` : the sdv table in one team's colours (title primary, line secondary); `team = NULL` : SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)` : `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight` : a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paleteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info` (`label = colour`), `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for **gt**: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the **reactable** helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column `to − from`: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "–"`, `share = FALSE`, `share_of = 1`, `share_location = "inline"` / `"column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: **Rolling-window wins ↗**

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars_top()` / `_bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: **Border bars ↗**

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column_subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: **Headers, legends, captions ↗**

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: **Faceted tables ↗** · **Schedule matrix ↗**

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir, match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: **Saving and posting ↗** · **Social posting ↗**

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv_logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv_wordmarks()`	the same
`reactable_sdv_headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv_team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv_team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv_cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv_theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: **reactable integration ↗**

SIBLINGS

Python **sdvplot** (PyPI) and JavaScript **@sportsdataverse/sdvplot** + **sdvtables** (npm) share sdvplotR's team index, logo archive and argument names; images render through **ggpath**.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in **ggplot2** plots, **gt** and **reactable** tables, and **plotly**, **ggiraph** and **leaflet**, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: **Getting started ↗**

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: **Logos, wordmarks, headshots ↗**

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE` `HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL` `SD` `OAK`; `SA` `TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: **Team colours and themes ↗**

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: **Logo axes and titles ↗**

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

`sdvplotR v0.3.0`

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league. **Football** `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x` `court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x` `pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for **gt**, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0
gt ≥ 0.10 · paletteer optional
GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()`: page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)`: `aer` `barttorvik` `basketball_reference` `bbc_sport` `bloomberg_markets` `cleaning_the_glass` `datagolf` `dunks_and_threes` `economist` `espn_site` `fivethirtyeight_site` `formula1` `gov_uk` `guardian` `nzz` `opta_analyst` `our_world_in_data` `reuters_graphics` `washington_post` `wikipedia`.

Cookbook: Table theme registry ↗

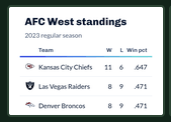
EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

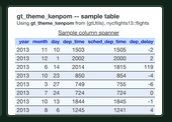
```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```

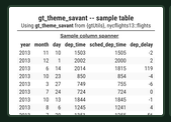


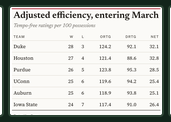
sdv



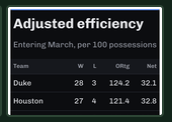
sdv · dark


kenpom


savant



broadsheet



midnight

`gt_theme_sdv()`: Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"`: on the SDV navy.

`gt_theme_sdv_team()`: the sdv table in one team's colours (title primary, line secondary); `team = NULL`: SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)`: `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight`: a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paletteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info` (`label = colour`), `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for **gt**: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the **reactable** helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column to − from: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "-"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: **Rolling-window wins ↗**

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars_top()` / `_bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: **Border bars ↗**

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column_subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: **Headers, legends, captions ↗**

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: **Faceted tables ↗** · **Schedule matrix ↗**

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir`, `match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: **Saving and posting ↗** · **Social posting ↗**

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv_logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv_wordmarks()`	the same
`reactable_sdv_headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv_team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv_team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv_cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv_theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: **reactable integration ↗**

SIBLINGS

Python **sdvplot** (PyPI) and JavaScript **@sportsdataverse/sdvplot** + **sdvtables** (npm) share sdvplotR's team index, logo archive and argument names; images render through **ggpath**.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in ggplot2 plots, gt and reactable tables, and plotly, ggiraph and leaflet, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: Getting started ↗

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: Logos, wordmarks, headshots ↗

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL SD OAK`; `SA TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: Team colours and themes ↗

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: Logo axes and titles ↗

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

`sdvplotR v0.3.0`

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league. **Football** `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for gt, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0

gt ≥ 0.10 · paletteer optional

GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()` : page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)` : `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

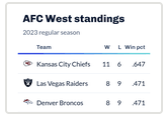
EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

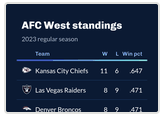
Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

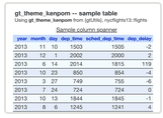
```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```

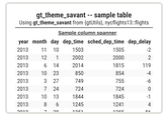


sdv



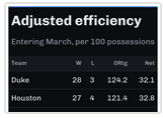
sdv · dark


kenpom


savant



broadsheet



midnight

`gt_theme_sdv()` : Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"` : on the SDV navy.

`gt_theme_sdv_team()` : the sdv table in one team's colours (title primary, line secondary); `team = NULL` : SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)` : `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight` : a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paleteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps"` `"blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info` (`label = colour`), `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for **gt**: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the **reactable** helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column `to − from`: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "–"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: Rolling-window wins ↗

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars _top() / _bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: Border bars ↗

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column _subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: Headers, legends, captions ↗

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: Faceted tables ↗ · Schedule matrix ↗

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir, match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: Saving and posting ↗ · Social posting ↗

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv _logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv _wordmarks()`	the same
`reactable_sdv _headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv _team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv _team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv _cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv _theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: reactable integration ↗

SIBLINGS

Python **sdvplot** (PyPI) and JavaScript **@sportsdataverse/sdvplot** + **sdvtables** (npm) share sdvplotR's team index, logo archive and argument names; images render through **ggpath**.

sdvplotR :: CHEAT SHEET

Team logos, wordmarks, player headshots and team colours for **28 leagues** (pro, college, minor and junior hockey, spring football, soccer, cricket) in ggplot2 plots, gt and reactable tables, and plotly, ggiraph and leaflet, behind one `league` argument. Built on ggpath, after the approach of nflplotR, cfbplotR, nbaplotR and mlbplotR. Page 1: ggplot2 marks, scales and axes. Page 2: teams, interactive, surfaces, coordinates. Page 3: gt images, themes, colour. Page 4: gt layout, saving, reactable.

sdvplotR v0.3.0

R ≥ 4.1 · ggplot2 ≥ 4.0 · ggpath ≥ 1.1

GALLERY ↗ COOKBOOKS ↗

GGPLOT2 · MARKS & SCALES

INSTALL & LOAD

```
install.packages("sdvplotR", repos = c(
  "https://sportsdataverse.r-universe.dev",
  "https://cloud.r-project.org"))
pak::pak("sportsdataverse/sdvplotR") # GitHub
library(sdvplotR); library(ggplot2)
```

Every team helper takes `league`, a key from `sdv_leagues()` (`"nfl"` `"nba"` `"cfb"` `"nhl"` `"ahl"` `"soccer"` …, page 2). Teams: any abbreviation, name or id `sdv_resolve()` reads (ESPN, nflverse, nba_api, MLB Stats, CFBD, …), or a conference (`"SEC"`, `"B1G"`, `"AFC"`).

Cookbook: Getting started ↗

LOGOS, WORDMARKS & HEADSHOTS

Geom	Required aes
`geom_sdv_logos()`	`x y team`
`geom_sdv_wordmarks()`	`x y team` (NFL, MLB, MiLB)
`geom_sdv_headshots()`	`x y player_id`
`geom_from_path()`	`x y path` (any image)

Optional aes and defaults: `alpha`, `colour` (`"b/w"` greyscales), `angle = 0`, `hjust = 0.5`, `vjust = 0.5`, `width = 1`, `height = 1` (npc; a typical logo is `width = 0.075`). Logos and wordmarks also take `season` (an aes or a constant).

```
df <- data.frame(
  team = sdv_teams("nba")$abbr,
  a = rnorm(30), b = rnorm(30))
ggplot(df, aes(a, b)) +
  geom_mean_lines(aes(x0 = a, y0 = b)) +
  geom_sdv_logos(aes(team = team),
                 league = "nba", width = 0.07)
```

HIGHLIGHT A GROUP

```
nfl <- subset(sdv_example_standings,
              league == "nfl")
west <- c("DEN", "KC", "LAC", "LV")
ggplot(nfl, aes(points_for,
                points_against)) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, !team %in% west),
    league = "nfl", width = 0.05,
    alpha = 0.25) +
  geom_sdv_logos(aes(team = team),
    data = subset(nfl, team %in% west),
    league = "nfl", width = 0.1)
```

Rows draw in order, so the last layer and row end on top. `colour = "b/w"` greys logos instead of fading them, which reads better on a dark background.

Cookbook: Logos, wordmarks, headshots ↗

GEOM ARGUMENTS

```
geom_sdv_logos(mapping = NULL,
  data = NULL, stat = "identity",
  position = "identity", ..., league,
  id_system = "auto", strict = FALSE,
  na.rm = FALSE, show.legend = FALSE,
  inherit.aes = TRUE)
geom_sdv_headshots(..., league,
  id_system = "espn")
```

`geom_sdv_wordmarks()` takes the logo arguments; headshots take no `season` or `strict`. Headshot `id_system`: `"espn"` = ESPN athlete ids in every league; `"gsis"` = nflverse NFL ids; `"league"` = NBA / WNBA Stats `PERSON_ID`, MLBAM, NHL API.

The ggproto objects `GeomSDVlogo`, `GeomSDVwordmark`, `GeomSDVheadshot` and ggpath's `GeomFromPath`, `GeomRefLines` are exported for extensions.

ERA LOGOS

```
old <- data.frame(x = 1:4, y = 1,
  team = c("QUE", "HFD", "ATL", "WPG"),
  season = c(1990, 1995, 2005, 2015))
ggplot(old, aes(x, y)) +
  geom_sdv_logos(aes(team = team,
    season = season), league = "nhl",
    width = 0.2) + theme_void()
```

`season`: the mark worn that year (ending year for split seasons); it also picks who held a reused code. NHL: every identity since 1917-18 (`QUE` `HFD`); AHL, ECHL, OHL, QMJHL, WHL, USHL, PWHL, PHF, UFL, XFL, AAF: dated marks; NFL / WNBA: relocated identities (`STL` `SD` `OAK`; `SA` `TUL`). Elsewhere today's mark. NHL marks are SVG: needs `rsvg`.

REFERENCE LINES

```
p + geom_mean_lines(aes(x0 = a, y0 = b))
p + geom_median_lines(aes(x0 = a),
  colour = "grey40", linetype = 1)
```

From ggpath. Optional aes `x0`, `y0`; defaults `colour = "red"`, `linewidth = 0.5`, `linetype = 2` (dashed), `alpha = NA`.

ANY IMAGE: GEOM_FROM_PATH

```
df$url <- sdv_logo_url(df$team, "nba",
  variant = "dark")
ggplot(df, aes(a, b)) +
  geom_from_path(aes(path = url),
                 width = 0.07)
```

`path` is a URL or a local file; same optional aes as the team geoms. Conferences resolve by key or other name: `sdv_logo_url(c("SEC", "B1G"), "cfb")`; a team's code wins (cfb `"MAC"` is Macalester).

COLOUR & FILL SCALES

```
scale_color_sdv(league, which = "primary",
  values = NULL, ..., aesthetics = "colour",
  breaks = waiver(), na.value = "grey50",
  guide = NULL, alpha = NA)
scale_fill_sdv(...)  # aesthetics = "fill"
```

Maps team keys to official colours; `which = "secondary"` for the second colour, `alpha` fades them (`scales::alpha()`), `values` replaces them, `...` goes to `scale_colour_manual()`. `scale_colour_sdv()` is an alias.

```
ggplot(df, aes(team, b)) +
  geom_col(aes(colour = team, fill = team)) +
  scale_color_sdv("nba", "secondary") +
  scale_fill_sdv("nba", alpha = 0.4)
```



NFL primary / secondary, as `sdv_team_colors("nfl", which = "all")` returns them.

Cookbook: Team colours and themes ↗

TITLES

```
p + ggtitle_image(title_image = "SEA",
  title = "Seattle Seahawks Analysis",
  image_height = 20, image_side = "left",
  league = "nfl") +
  theme_title_image(size = 20, hjust = 0.5)
```

`title_image`: a team key (logo resolved) or any image URL. `image_height = 15` px by default; `image_side = "left"` or `"right"`; `subtitle`. `theme_title_image(...)` is required and passes `...` to `ggtext::element_markdown()`.

LOGO AT EACH BAR'S END

```
nba <- subset(sdv_example_standings,
              league == "nba")
ggplot(nba, aes(wins,
    reorder(team, wins))) +
  geom_col(aes(fill = team)) +
  geom_sdv_logos(aes(x = wins + 3,
    team = team), league = "nba",
    height = 0.028) +
  scale_fill_sdv("nba")
```

`height` sizes the logo here: each row is about a thirtieth of the panel.

LOGO AXES

Route	Calls
theme element	`element_sdv_logo()` · `element_sdv_wordmark()` · `element_sdv_headshot()`
scale + theme	`scale_x_sdv()` / `scale_y_sdv()` · `scale_x_sdv_headshots()` + `theme_x_sdv()` / `theme_y_sdv()`

```
# element route (size in cm)
ggplot(df, aes(team, b)) + geom_col() +
  theme(axis.text.x =
    element_sdv_logo("nba", size = 1))
```

`element_sdv_logo(league, alpha = NULL, colour = NA, hjust = NULL, vjust = NULL, size = 0.5, season, id_system)`; `_wordmark()` the same; `_headshot()` takes `id_system`, no `season`.

```
# scale route (size in px; needs ggtext)
ggplot(df, aes(b, team)) + geom_col() +
  scale_y_sdv(league = "nba", size = 12) +
  theme_y_sdv()
```

`scale_x_sdv(..., league, expand, guide, position = "bottom", size = 12, season, id_system)`; `scale_y_sdv()` has `position = "left"`. Size is the height on x, the width on y. `scale_x_sdv_headshots()` (`size = 20`) and `scale_y_sdv_headshots()` (`size = 30`) take `id_system`. `theme_x_sdv()` / `theme_y_sdv()` render those labels as markdown.

Cookbook: Logo axes and titles ↗

HEADSHOTS ON A SCATTER

```
ggplot(scorers, aes(ppg, ts)) +
  geom_sdv_headshots(aes(player_id =
    athlete_id), league = "nba",
    height = 0.12)
```

ESPN ids: the `athlete_id` of hoopR, wehoop, cfbfastR rosters. NFL gsis ids (`id_system = "gsis"`) go through a headshot map built from nflverse rosters (1999 on); `sdv_clear_cache()` re-reads it.

IMAGES IN THEME ELEMENTS

```
p + theme(plot.background =
  element_sdv_raster("court.png"))
```

`element_sdv_raster(image_path, x, y, width, height, just = "centre", hjust = 0.5, vjust = 0.5, interpolate = TRUE, ...)`: one image in a rectangle; wraps ggpath's `element_raster()` (also exported).

`element_path(alpha = 1, colour = NA, hjust, vjust, angle = 0, size = unit(0.5, "cm"))`: axis text that already holds image paths or URLs.

sdvplotR :: TEAMS, INTERACTIVE & SURFACES

The shared SportsDataverse team index behind every helper: **28 leagues**, any team identifier resolved to one canonical `team_id`, colours, palettes and archived logo URLs. Plus logos on **plotly**, **ggiraph** and **leaflet**, premade tier plots, 59 playing surfaces drawn with **sportyR** (team-coloured in the team leagues), and converters that put shot and event coordinates from stats.nba.com, Euroleague and ten soccer providers on one frame.

`sdvplotR v0.3.0`

sportyR ≥ 2.4.5 for surfaces · rsvg for SVG logos

GALLERY ↗ COOKBOOKS ↗

TEAM DATA · LEAGUES, KEYS, COLOURS

LEAGUES & TEAMS

```
sdv_leagues()  # the 28 league keys
sdv_teams("nfl")
sdv_teams("cfb", include_conferences = TRUE)
```

Group	`sdv_teams()` columns
ids, names	`league team_id abbr name short_name location`
grouping	`program conference_id conference`
colours	`color_primary color_secondary color_source`

`team_id` is the canonical key (the ESPN id where ESPN has one). Conference rows (`"SEC"`, `"AFC"`) draw logos only.

RESOLVE ANY IDENTIFIER

```
sdv_resolve(c("GSW", "Golden State Warriors",
  "XYZ"), "nba")
#> "9" "9" NA   + one warning
sdv_resolve(c("OAK", "SD", "STL"), "nfl")
#> "13" "24" "14"   LV, LAC, LA
sdv_suggest("Kansas Cty Chiefs", "nfl")
```

`sdv_resolve(x, league, season = NULL, id_system = "auto", strict = FALSE)` reads abbreviations, names and ids from ESPN, nflverse, MLB Stats, nba_api, NHL, HockeyTech, NCAA, PFF, CFBD, Sports Reference, FanGraphs and more. `id_system` names one (`"nhl_id"` only explicitly). `season` picks who held a reused code (ending year for split seasons). Unresolved: `NA` and one warning (`strict = TRUE` errors). `sdv_suggest(x, league, n = 5)` lists candidates, never picks.

Every helper that takes a team resolves it this way, with `league` and `id_system`. `strict`: the logo-URL, colour, tier and mark-drawing helpers; not `sdv_marks()`, `element_sdv_*()`, `scale_x/y_sdv()`, `sdv_colorway()`, `sdv_team_factor()`.

COVERAGE

Leagues	Teams
nfl nba mlb nhl wnba	32 · 30 · 30 · 32 · 15
nbagl milb pwhl phf	32 · 286 · 12 · 8
ufl xfl usfl aaf	11 · 11 · 9 · 8
ahl echl ohl whl qmjhl ushl	61 · 59 · 27 · 23 · 30 · 17
cfb mbb wbb	687 · 366 · 364
ncaa_ baseball softball mhockey whockey	414 · 415 · 110 · 48
soccer · cricket	2,631 · 111

Team rows in the bundled index `0aeef853ec2b`; the live index can differ.

COLOURS & PALETTES

```
sdv_team_colors("nfl", "KC")
sdv_team_colors("nfl", c("KC", "BUF"),
                which = "all")
sdv_team_colors("nba", which = "all")
sdv_color_palette("nba",
  teams = c("BOS", "NY"))
```

`sdv_team_colors(league, team = NULL, which = "primary", season, id_system, strict, include_conferences = FALSE)`: a named hex vector (`NA` where a team does not resolve); `which = "all"` gives `"primary, secondary"` strings, or with `team = NULL` a data frame `team_id abbr name primary secondary`. `sdv_color_palette(league, teams = NULL, which = "primary", ...)`: named colours for `scale_*_manual()`.

IMAGE URLS & MARKS

```
sdv_logo_url(c("KC", "Buffalo Bills",
  "OAK"), "nfl")
sdv_logo_url("NYJ", "nfl",
  variant = "scoreboard")
sdv_logo_url("QUE", "nhl", season = 1990)
sdv_logo_url("KC", "nfl",
  mark_type = "wordmark")
sdv_marks("KC", "nfl")  # every mark
sdv_headshot_url("2544", "nba",
  id_system = "league")
```

`sdv_logo_url(team, league, season = NULL, variant = "default", mark_type = "logo", id_system, strict)`: the SportsDataverse archive's immutable copy. `variant`: `"default"`, `"dark"`, or a name from `sdv_marks()` (`"scoreboard"`, NFL `"grayscale"`, MLB `"cap_on_light"`, …); a team without it gets its closest mark of the same polarity, an unknown name errors. `mark_type = "wordmark"`: NFL, MLB (and MiLB).

`sdv_headshot_url(player_id, league, id_system = "espn")`: ESPN headshots for nfl nba wnba mlb nhl cfb mbb wbb; `"gsis"` and `"league"` as on page 1. `NA` for a missing or malformed id.

CACHE & VERSIONS

Index and logo manifest: SportsDataverse CDN, cached a day; a bundled copy serves `options(sdvplotR.offline = TRUE)`. `sdv_versions()`: what is loaded. `sdv_clear_cache()`: drop index, manifest, headshot map, images.

PREMADE PLOT: TEAM TIERS

```
tiers <- data.frame(
  tier_no = c(1, 1, 2, 2, 3),
  team = c("KC", "DET", "BAL", "PHI", "BUF"))
sdv_team_tiers(tiers, league = "nfl",
  theme = "light", no_line_below_tier = 2)
```

`sdv_team_tiers(data, league, title, subtitle, caption = NULL, tier_desc, presort = FALSE, alpha = 0.8, width = 0.075, no_line_below_tier = NULL, devel = FALSE, theme = "dark", variant = "auto", season, id_system, strict)`. `data` needs `tier_no` and `team` (optional `tier_rank`). Title `"NFL Team Tiers"` from `league`; `NULL` drops one. `devel = TRUE` plots keys, not logos; `"auto"`: dark marks on the dark theme.

Cookbook: College hoops tiers ↗

PLAYING SURFACES

```
sdv_surface("nba", "BOS",
            center_logo = TRUE) +
  geom_point(aes(x, y), data = shots)
```

59 keys, each a sportyR `geom_<sport>()` of that league.
Football `nfl aaf ufl usfl xfl cfb cfl nfhs_football` · **basketball** `nba nbagl wnba mbb wbb fiba nfhs_basketball` · **baseball** `mlb milb pony ncaa_baseball little_league nfhs_baseball` · **softball** `ncaa_softball` · **hockey** `nhl whl pwhl ahl echl ohl qmjhl ushl phf iihf nwhl ncaa_mhockey ncaa_whockey` · **soccer** `soccer fifa epl mls nwsl ncaa_soccer` · **lacrosse** `nll pll world_lacrosse ncaa_mlacrosse ncaa_wlacrosse usa_mlacrosse usa_wlacrosse` · **tennis** `atp wta itf ita usta ncaa_tennis` · **volleyball** `fivb usa_volleyball ncaa_volleyball` · **curling** `wcf curling_canada`. `whl` uses the NHL rink; PWHL and G League draw their own.

`sdv_surface(league, team = NULL, season = NULL, center_logo = FALSE, ...)`. A `team` (team leagues only) paints the paint and apron, end zones, or centre line, circle and boards; soccer gets a centre logo only; a diamond stays regulation and `center_logo` goes in shallow centre field. Stylized colours, not real floors; origin at centre (home plate on a diamond), in feet (yards for football, metres for soccer and FIBA); `...` goes to the sportyR geom (`rotation`, `color_updates`).

Cookbook: Surfaces and shot charts ↗

TEAM FACTOR

```
sdv_team_factor(c("KC", "BUF", "invalid"),
                league = "nfl")
#> KC BUF <NA>   Levels: BUF KC
```

Levels by abbreviation; unresolved values become `NA`.

INTERACTIVE: GGIRAPH, PLOTLY, LEAFLET

```
p <- ggplot(d, aes(x, y)) +
  geom_sdv_logos_interactive(aes(team = team,
    tooltip = team, data_id = team),
    league = "nfl", width = 0.1)
ggiraph::girafe(ggobj = p)
plotly::plot_ly(d, x = ~x, y = ~y) |>
  add_sdv_logos(d$x, d$y, d$team, "nfl") |>
  sdv_colorway("nfl", d$team)
leaflet::leaflet(d) |> leaflet::addTiles() |>
  addSdvLogoMarkers(~lon, ~lat, ~team,
                    league = "nfl")
```

`geom_sdv_{logos,wordmarks,headshots}_interactive()`: images (at most `max_px = 160` a side) plus an invisible hover point (`hover_size = 8`). plotly: `add_sdv_logos(p, x, y, team, league, size = 0.08, ...)`, `add_sdv_wordmarks()`, `add_sdv_headshots()`, `sdv_axis_logos(p, teams, league, axis)` tick logos, `sdv_colorway()`; `ggplotly()` drops them. `addSdvLogoMarkers(..., size = 32)`: no logo keeps the default marker.

Cookbook: Interactive plots ↗ · leaflet maps ↗

SHOT & EVENT COORDINATES

```
# stats.nba.com legacy frame -> feet
shots |> sdv_court_coords()
# Euroleague (cm, hoop origin) -> FIBA m
pts |> sdv_court_coords(
  x_column = "coord_x", y_column = "coord_y",
  provider = "euroleague")
# soccer events -> 105 x 68 m pitch
ev |> sdv_pitch_coords("statsbomb")
```

`sdv_court_coords(data, x_column = "x_legacy", y_column = "y_legacy", provider = "nba")` adds `court_x court_y`: basket at −41.75 ft (−12.425 m for Euroleague), half court at 0; `court_y` −25 to 25 ft, negative = shooter's left. From hoopR's `nba_shotchartdetail()`: `sdv_court_coords(res$Shot_Chart_Detail, "LOC_X", "LOC_Y")`.

`sdv_pitch_coords(data, provider, x_column = NULL, y_column = NULL, flip = NULL, pitch_length = NULL, pitch_width = NULL)` adds `pitch_x pitch_y` in metres from the centre spot (`pitch_x` −52.5 = defended goal line). `flip`: `TRUE` / `FALSE` or a logical column, as `flip = "away"`, turns rows half a turn. `pitch_length` / `pitch_width`: tracking only. Providers: `opta` (`statsperform`) `wyscout statsbomb uefa impect espn`, tracking `tracab skillcorner secondspectrum metrica`.

Cookbook: NHL shot map ↗ · Soccer ↗

sdvplotR :: gt TABLES

Team images, colours and house themes for gt, **122 surveyed table styles**, plus the **gtUtils** toolkit (by Andrew Weatherman): 18 more themes, legends, colouring, annotations, layouts and image export. Every helper takes a `gt_tbl` as its first argument (`gt_object`) and returns one, so they chain with `|>`. Examples run on `sdv_example_standings` (real NFL and NBA final standings). Page 3: images, themes, colour, legends, focus. Page 4: format, layout, saving, reactable.

sdvplotR v0.3.0
gt ≥ 0.10 · paletteer optional
GALLERY ↗ COOKBOOKS ↗

GT · IMAGES, THEMES, COLOUR

TEAM IMAGES IN CELLS

```
gt_sdv_logos(gt_object, columns, league,
  height = 30, locations = NULL,
  include_name = FALSE, season = NULL,
  id_system = "auto", strict = FALSE)
gt_sdv_wordmarks(gt_object, columns,
  league, height = 30, locations = NULL,
  season = NULL, id_system, strict)
gt_sdv_headshots(gt_object, columns,
  league, height = 30, locations = NULL,
  id_system = "espn")
gt_sdv_cols_label(gt_object,
  columns = everything(), league,
  height = 30, mark_type = "logo",
  id_system, season, strict)
```

`height` in px. `locations` (`cells_body()`, `cells_stub()`, `cells_column_labels()`, `cells_row_groups()`) replaces `columns`. `include_name = TRUE` keeps the text after the logo; `season` shows one season's marks in every cell. Wordmarks: NFL, MLB (and MiLB). `gt_sdv_cols_label()` turns column *labels* into images: `mark_type = "logo" "wordmark" "headshot"`.

STACKED NAME IN TEAM COLOUR

```
gt_merge_stack_team_color(gt_object,
  col1, col2, team_col, league,
  font_size_top = 14,
  font_size_bottom = 12, color = "black",
  background = NULL, season, id_system)
```

`col1` in bold caps over `col2` in the team's colour; a primary under 4.5:1 contrast gives way to the secondary or darkens. Theme first: `background = NULL` reads it.

SURVEYED TABLE STYLES

```
sdv_table_themes(category = "sports")
g <- gt(head(sdv_example_standings))
g |> gt_theme_source("datagolf", "dark")
g |> gt_theme_wikipedia()
```

122 source-named styles in 7 categories (98 with a dark mode), shared with sdvplot and sdvtables.
`table_theme_spec()` resolves one;
`reactable_sdv_theme()`: page 4. 20 shortcuts
`gt_theme_<id>(gt_object, mode, density)`: `aer barttorvik basketball_reference bbc_sport bloomberg_markets cleaning_the_glass datagolf dunks_and_threes economist espn_site fivethirtyeight_site formula1 gov_uk guardian nzz opta_analyst our_world_in_data reuters_graphics washington_post wikipedia`.

Cookbook: Table theme registry ↗

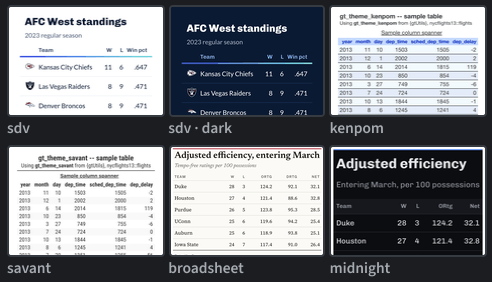
EXAMPLE

```
library(gt)
nfc <- subset(sdv_example_standings,
  division == "NFC North",
  c(team, team_name, wins, losses,
    last_season_wins))
gt(nfc) |>
  gt_sdv_logos(columns = team,
               league = "nfl", height = 28)
```

Cookbook: Team tables with the gt toolkit ↗

HOUSE & GTUTILS THEMES

```
gt_theme_sdv(gt, style = "light")
gt_theme_sdv_team(gt, team = "KC",
                  league = "nfl")
```



`gt_theme_sdv()`: Chivo labels, a Lato body in tabular figures, one SDV-gradient horizon line (`#3346F0` → `#7FE6DC`); `style = "dark"`: on the SDV navy.

`gt_theme_sdv_team()`: the sdv table in one team's colours (title primary, line secondary); `team = NULL`: SDV colours; takes `season`, `id_system`.

These 20 `gt_theme_<name>(gt_object, ..., density)` take `density = "comfortable" "compact" "social"` (default comfortable; compact for almanac, scoreboard, terminal) and pass `...` to `gt::tab_options()`, applied last.

Option	Themes (default)
`accent`	almanac `"#8C2F1E"` · booktabs, swiss, tufte `"#111111"` · broadsheet `"#A6081A"` · brutalist `"#FF3B00"` · midnight `"#5B8DEF"` · scoreboard `"#0E1621"` · terminal `"#FFB86C"`
`style`	sdv, sofa `"light"` / `"dark"` · tier `"dark"` / `"light"`
other	drench `color = "#123F5E"` · broadsheet `paper = "white"` or `"salmon"` · almanac `stripe` (`NA` = off)
none	athletic gtutils kenpom ncaa pl savant

Cookbook: SportsDataverse table themes ↗

PREVIEW & PALETTES

```
gt_theme_preview(nfc, themes = c(
  "gt_theme_sdv", "gt_theme_kenpom"))
gt_color_ranks(..., palette = pal_midnight)
```

`gt_theme_preview(data, themes = NULL, n = 5, ncol = 3, density = "compact", file = NULL, ...)`: `NULL` shows every theme; `themes` needs full function names. `...` goes to `gt_grid()`.

`pal_midnight`: a rank palette that reads on `midnight` / `terminal`.

COLOUR BY VALUE

Function	Key arguments
`gt_color_ranks()`	fill cells: `palette` `domain` `reverse`, `na_color = "white"`, `autocolor_text = TRUE`
`gt_color_pills()`	pill: `fill_type = "continuous"` / `"rank"`, `format_type`, `suffix`, `digits`
`gt_percentile_bar()`	bar + marker: `domain = c(0, 100)`, `scale = "auto"` (0–1 → 0–100), `na_label = "—"`
`gt_color_results()`	row colours: `result_column = "result"`, `result_type = "wl"` / `"binary"`
`gt_row_accent()`	edge bar: `column`, `palette`, `side = "left"`, `hide = TRUE`
`gt_indicator_boxes()`	boxes: `indicator_vals = c(0, 1)`, `color_yes`, `show_text`
`gt_tiers()`	tier list: `levels`, `colors`, `tier_column = "tier"`, `style = "dark"`

Most take `columns` and `rows` (an expression such as `wins > 10`, or indices). `palette` also takes a paletteer name, e.g. `"viridis::mako"`; `domain = NULL` uses the data's range.

Cookbook: Percentile bars and cut lines ↗

TIER LIST

```
gt(seeds) |>  # tier + logo URL columns
  gt_tiers(levels = c("Bye", "Wild card"),
    colors = c("#1B7837", "#B2182B")) |>
  gt_legend_discrete()
```

Columns other than `tier_column` render as images (`image_columns` narrows them; `img_height = "55px"`), e.g. a column of `sdv_logo_url()` URLs. `levels` may be one named vector, `c(Bye = "#1B7837")`.

Cookbook: Tier lists ↗

LEGENDS

Function	Key arguments
`gt_legend_continuous()`	bar: `type = "continuous" "steps" "blocks"`, `n_bins = 5`, `labels ("edges")`, `title`, `location = "bottom"`, `width = 200`
`gt_legend_discrete()`	swatches: `key_info` (`label = colour`), `heading`, `location = "top"`, `shape`, `direction`

```
gt(nfc) |>
  gt_color_ranks(wins) |>
  gt_legend_continuous(type = "steps",
                       title = "Wins")
```

The continuous legend reads the columns, palette and domain the colouring step recorded; the discrete one reads the key `gt_tiers()` recorded, so `key_info` can be left out.

```
gt(nfc) |>
  gt_legend_discrete(c("Playoffs" =
    "#CCE7F5", "Missed" = "#F5CCCC"))
```

Cookbook: Headers, legends, captions ↗

HIGHLIGHT & FOCUS

Function	Key arguments
`gt_highlight_cells()`	`columns`, `condition` (`~ .x > 0.7` or a logical mask), `fill = "#FFF3B0"`
`gt_highlight_na()`	`fill = "#F0F0F0"`, `missing_text`, `na_strings = "NA"`
`gt_outliers()`	`method = "iqr" "sd" "bounds"`, `threshold` (1.5 / 3), `side`, `symbol`, `note`
`gt_spotlight()`	`rows`; dims the rest (`dim_color = "auto"`, WCAG AA); `accent_color`, `if_none`
`gt_bold_rows()`	`rows`, `text_color = "black"`, `highlight_color`
`gt_group_stripes()`	band row groups: `color = "#F5F5F5"`, `start = 2`

```
gt(nfc) |>
  gt_spotlight(team == "DET",
               accent_color = "#0076B6") |>
  gt_outliers(wins, method = "sd",
              note = TRUE)
gt(nfc) |>
  gt_highlight_cells(c(wins, losses),
    ~ .x >= 10, bold = TRUE)
```

sdvplotR :: gt LAYOUT & reactable

The rest of the gtUtils toolkit for gt: number formats, rules and border bars, headers and captions, multi-table layouts and image export for posting. Then the reactable helpers: team logos, wordmarks and headshots in cells and headers, team-coloured bars and fills, and registry themes. Examples continue with `nfc` (the NFC North rows of `sdv_example_standings`, page 3).

sdvplotR v0.3.0

webshot2 to save · reactable for tables

GALLERY ↗ COOKBOOKS ↗

GT · FORMAT, LAYOUT, SAVE

FORMAT

Function	Key arguments
`gt_delta()`	change column to − from: `column_label = "Change"`, `percent = FALSE`, `decimals = 1`, `arrows = FALSE`, `after`
`gt_fmt_rank()`	1 → 1st: `superscript = TRUE`, `suffix_size = "0.7em"`
`gt_fmt_tally()`	W-L(-T) in one cell: `separator = "-"`, `share = FALSE`, `share_of = 1`, `share_location = "inline" / "column"`
`gt_significance()`	stars from `p_columns`: `levels = c(0.01, 0.05, 0.1)`, `symbols`, `hide_p = TRUE`
`gt_scale_note()`	divide and say so: `divisor = 1000` ("Figures in thousands."), `where = "source_note" "label" "both"`
`gt_wrap_labels()`	break long labels: `width = 12`, `balance = TRUE`
`gt_set_font()`	one family for the table: `font_family`, `from_google_font = TRUE`, `weight`, `style`

```
gt(nfc) |>
  gt_fmt_tally(c(wins, losses)) |>
  gt_delta(from = last_season_wins,
           to = wins)
```

Cookbook: Rolling-window wins ↗

BORDERS & RULES

Function	Key arguments
`gt_border_grid()`	full grid: `color = "black"`, `weight = 1`, `include_labels = FALSE`
`gt_border_bars_top()` / `_bottom()`	colour bars: `colors`, `bar_height = 10`, `img`, `text`, `text_size = 18`
`gt_cutline()`	line after rows: `after`, `label`, `color = "#A6081A"`, `style = "dashed"`, `gap`

```
gt(nfc) |>
  gt_border_bars_top(
    c("#0076B6", "#B0B7BC")) |>
  gt_cutline(after = c(1, 3),
    label = c("Champion", NA))
```

Cookbook: Border bars ↗

HEADERS & ANNOTATIONS

Function	Key arguments
`gt_title_header()`	`title`, `subtitle`, `kicker`, `date`, one `*_style` list each
`gt_column_subheaders()`	per column `list(heading, subtitle)` in `...`; `subtitle_color = "#808080"`
`gt_538_caption()`	`top_caption` over a rule, `bottom_caption`, `align = "right"`
`gt_marginalia()`	a column as notes: `width = 220`, `italic = TRUE`, `rule = TRUE`
`gt_social_tag()`	handles + icons: `accounts`, `caption`, `stack = FALSE`, `separator = " | "`
`gt_watermark()`	`text` or `image`, `opacity = 0.06`, `size = "60%"`, `angle`

```
gt(nfc) |>
  gt_title_header("NFC North",
    kicker = "2025 SEASON",
    subtitle = "Final standings") |>
  gt_column_subheaders(wins = list(
    heading = "W", subtitle = "2025")) |>
  gt_social_tag(c(x = "@SportsDataverse",
                  gh = "sportsdataverse"))
```

`accounts` names map to Font Awesome brand icons, with aliases (`x` / `twitter`, `ig`, `bsky`, `gh`). A `Date` in `date` prints as "July 21, 2026".

Cookbook: Headers, legends, captions ↗

SNAKE A LONG TABLE

```
nba <- subset(sdv_example_standings,
  league == "nba", c(team, wins, losses))
gt(nba) |> gt_snake(n_cols = 3)
```

Thirty rows become three blocks of ten, each with its own labels. Body-cell styling applied before `gt_snake()` carries through; formats and content transforms (such as `gt_sdv_logos()`) go after it.

LEARN MORE

GALLERY ↗ every chart and table in one place

Start: Getting started · **All:** articles · reference

Per league: NFL · CFB · NBA · WNBA · MLB · NHL · MBB · WBB · Soccer

Recipes: NFL EPA · CFB table · WBB standings · head-to-head card

New: table themes · reactable · leaflet maps

Leaderboards: dashboards with Quarto · **Automation:** social graphics

LAYOUT

Function	Key arguments
`gt_grid()`	tables in a grid: `tables`, `ncol = 2`, `labels`, `title`, `caption`, `gap = 24`, `align = "top"`, `file`
`gt_stack_tables()`	one above another: `gap = 16`, `align = "center"`, `title`, `file`
`gt_snake()`	long table in blocks: `n_cols = 2` or `rows_per_col`, `gap = 20`, `fill = ""`
`gt_snake_align()`	reshape a parallel frame (a mask for `gt_highlight_cells()`) to match

```
gt_grid(list(afc_tbl, nfc_tbl),
        ncol = 2, title = "2025 standings",
        file = "standings.png")
```

Grids and stacks share `subtitle`, `source_note`, `caption_rule`, the `*_style` lists, and the save arguments `bg = "white"`, `whitespace = 50`, `zoom = 2`.

Cookbook: Faceted tables ↗ · Schedule matrix ↗

SAVE AS AN IMAGE

Function	Key arguments
`gt_save_crop()`	PNG / JPG, trimmed and padded: `file = NULL`, `bg = "white"`, `whitespace = 50`, `zoom = 2`, `expand = 5`, `width`
`gt_social_crop()`	pad to `aspect_ratio = "1:1"`, `"16:9"`, `"4:5"` or a number; `whitespace = 60`, `gravity = "center"`
`gt_save_batch()`	one image per group: `data, group, fn, file, dir, match_width = TRUE`

```
tbl |> gt_save_crop("table.png",
                   whitespace = 40,
                   width = 1200)
# one image per conference
nfl <- subset(sdv_example_standings,
              league == "nfl")
build <- function(df, group)
  gt(df) |> tab_header(title = group)
gt_save_batch(nfl, conference, build,
  "nfl-{group}.png", dir = "out")
```

Saving runs headless Chrome through `webshot2` (Chrome gets 60 s to start). `file = NULL` returns the PNG bytes. `dir` has no default, so a batch never lands in the working directory unasked.

Cookbook: Saving and posting ↗ · Social posting ↗

REACTABLE

REACTABLE TABLES

Helper	Arguments (defaults)
`reactable_sdv_logos()`	`league`, `variant = "default"` (`"dark"`, a named variant), `height = 30`, `default_img = NULL`
`reactable_sdv_wordmarks()`	the same
`reactable_sdv_headshots()`	`league`, `height = 40`, `default_img`, `id_system = "espn"`
`reactable_sdv_team_color_bar()`	`data`, `team_col`, `league`, `which = "primary"`, `max_value = NULL`, `na_color = "grey70"`
`reactable_sdv_team_color_bg()`	as the bar, with `alpha = 0.15` in place of `max_value`
`reactable_sdv_cols_label()`	`.data`, `...` (to every `colDef()`), `league`, `variant`, `height = 30`
`reactable_sdv_theme()`	`name`, `mode = "light"`: a registry theme (page 3) for `reactable(theme = )`

```
library(reactable)
reactable(df, columns = list(
  team = colDef(html = TRUE,
    cell = reactable_sdv_logos("nfl")),
  wins = colDef(style =
    reactable_sdv_team_color_bar(
      df, "team", league = "nfl"))))
# columns named by team key -> logo headers
reactable(h2h,
  columns = reactable_sdv_cols_label(
    h2h, league = "nba"))
```

All but headshots also take `season`, `id_system`, `strict`. The first three are `cell` renderers (`colDef(html = TRUE)`); unresolved values show as text, or `default_img`; a variant that fails to load swaps to the default image. The colour helpers are `style` functions: the bar's length is the value over `max_value` (the column maximum by default). `reactable_sdv_cols_label()` leaves out columns it cannot resolve, so they keep their plain names.

Cookbook: reactable integration ↗

SIBLINGS

Python sdvplot (PyPI) and JavaScript @sportsdataverse/sdvplot + sdvtables (npm) share sdvplotR's team index, logo archive and argument names; images render through ggpath.